Independent review: exact mount-diff proposal

1 November 2026

The published pure parser passed 43 independent fixed cases: seven accepted inputs (six orderings plus the retained native diff), and 36 rejected malformed or incorrectly typed inputs. Every accepted result keeps writer attribution null and the no-writes claim false.

Exact proposed contract

Only three unique added entries are accepted: `/fixtures`, `/qualification`, and `/qualification/exec-gate`. Input must be exact bytes, ASCII, at most 1,024 bytes, with exactly one final newline. Changed/deleted entries, malformed framing, path normalization variants, duplicates and extras are rejected.

Retained qualification remains failed

The v5 receipt records Q0 passed and Q1 stopped on an unexpected writable-layer change; cleanup completed, qualification did not. Its native diff has the proposed three additions. Accepting those historical bytes under a different rule is a prospective amendment, not a passing reinterpretation of v5.

The retained Q1 create/terminal metadata have matching full identity, image, name, two expected read-only binds and read-only rootfs. This is consistency of retained metadata, not a current daemon observation. The original supervisor still requires an empty diff and does not call this parser.

Required amendment and adapter review

Version the new contract and bind parser, supervisor, review, image, gate, binaries and fixtures. Apply the revised predicate only after the existing exact identity, read-only mount, privilege, namespace and resource checks. Require successful diff transport, complete bounded raw bytes, parser success and exact-ID cleanup. Test the composed adapter against invalid identity/mounts and transport/parser failures; pure parser acceptance cannot replace prerequisites.

Retain Q0–Q4 requirements and the 300-second total, 30-second cleanup reserve and 20-second active-case bounds if execution is later admitted. This review supplies no deployment, runtime authorization or admission. The detailed amendment list is in `AMENDMENT_REQUIREMENTS.md`.

Evidence and interpretation

Controls cover ordering, incorrect types, missing/extra/duplicate changes, C/D operations, path variants, whitespace, NUL/non-ASCII and size limits. Source and retained receipt bindings were checked unchanged. No supervisor, Docker, container, solver or scientific workload was run.

Docker's primary CLI reference describes changes since creation and A/C/D status, not the writer or cause. Diff acceptance cannot establish absence of all writes; bound mounts and tmpfs need separate evidence. Historical pre-exec PID observations retain their stated temporal scope.

Source: PR3056 commit `05fc018e64e948fbe91d5463036e55d4c97d13f1`. Review was source-first, not blind; the reviewer did not author the parser. `status.json`, control receipts and hashes preserve the scoped result.